

FINAL PROJECT REPORT

Serverless MLOps: 3-Day Air Quality Index (AQI) Predictor

A multi-output regression system for 24h, 48h, and 72h
AQI forecasting using fully serverless infrastructure

Architecture: FastAPI · Streamlit · GitHub Actions

Models: RF · XGBoost · LightGBM · Ridge · LSTM

Data Sources: AQICN API · Open-Meteo API

Storage: Local Parquet Feature Store

Serverless · CI/CD Automated · Explainable AI

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1. Executive Summary & Problem Statement

Air pollution remains a critical global health crisis. Static index reporting tells citizens the air quality *right now*, but fails to provide the foresight required to plan outdoor activities, protect vulnerable demographics, or initiate local policy interventions.

This project delivers a **Serverless MLOps system** that predicts the AQI for the upcoming three days — at 24-hour, 48-hour, and 72-hour horizons — using a multi-output regression setup. By tracking ambient air pollutants and atmospheric conditions, the system captures complex temporal patterns.

System Highlights

- **Fully serverless** — zero infrastructure cost
- **GitHub Actions** for CI/CD automation
- **Local Parquet** database as feature store and model registry
- **FastAPI** backend for model serving
- **Streamlit** UI dashboard for interactive exploration

2. System Architecture & The 4-Pipeline Loop

The application is structured into four independent, decoupled pipelines. This decoupled nature ensures that ingestion, training, serving, and UI rendering can scale and fail independently.

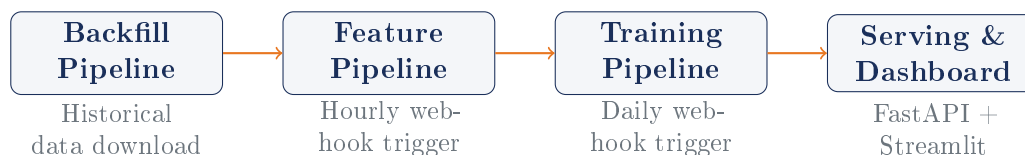


Figure 1: High-level view of the 4-pipeline MLOps loop.

2.1 Pipeline 1 — Backfill (Historical Data)

Downloads historical weather and air quality data from the Open-Meteo API. Engineers temporal features and shifts target columns (`aqi_next_24h`, `aqi_next_48h`, `aqi_next_72h`), saving the initial dataset into a local Parquet database (`data/local_feature_store/`).

2.2 Pipeline 2 — Feature Pipeline (Hourly Webhook)

Executed hourly on GitHub Actions compute runners. GitHub’s native `schedule` cron trigger is notoriously inconsistent — it can drift by 15–30 minutes or skip runs entirely under heavy load. To guarantee reliable, on-time execution, the pipeline is triggered by an **external webhook service** (`cron-job.org`), which sends an HTTP `POST` to the GitHub Actions webhook endpoint at precise intervals. GitHub’s own `schedule` trigger is kept enabled as a fallback, so both are active simultaneously.

Note on “Manual” Run Labels

Runs triggered via the external webhook API appear in the GitHub Actions UI as “**Manually run by *username***” rather than “Scheduled.” This is a GitHub labelling artefact — the underlying trigger is the external cron job firing automatically. It does *not* indicate a human pressed the button.

The pipeline fetches current pollutant statistics from the AQICN API, formats the data, appends the new row into the Parquet database, and automatically commits the updated database back to the GitHub repository.

2.3 Pipeline 3 — Training Pipeline (Daily Webhook)

Executed daily on GitHub Actions compute runners — also triggered externally via `cron-job.org` webhook for the same reliability reasons. GitHub’s own schedule is kept on as a fallback. It pulls the updated Parquet database, performs chronological time-series splitting, trains candidate regression models, and scores them. The best performing model artifacts and SHAP feature importance plots are saved to the `outputs/` folder and automatically committed back to the repository.

2.4 Pipeline 4 — Serving & Dashboard (FastAPI + Streamlit)

- **FastAPI Backend:** Exposes the `/predict` endpoint, which loads the latest trained models from the local registry, fetches live features, and serves the 3-day AQI forecast.
- **Streamlit Dashboard:** A user-friendly frontend for inspecting current air quality, exploring predictions, reading explainability plots, and interacting with data in real-time.

✓ Feature Pipeline (Hourly) Feature Pipeline (Hourly) #7: Manually run by Ateeb30	main	50 minutes ago 1m 47s	...
✓ Training Pipeline (Daily) Training Pipeline (Daily) #7: Scheduled	main	Today at 6:10 PM 2m 9s	...
✓ Feature Pipeline (Hourly) Feature Pipeline (Hourly) #6: Scheduled	main	Today at 6:09 PM 1m 35s	...
✓ Feature Pipeline (Hourly) Feature Pipeline (Hourly) #5: Manually run by Ateeb30	main	Today at 6:00 PM 1m 35s	...

Figure 2: GitHub Actions tab showing automated pipeline runs. Runs labelled “Manually run” are in fact triggered automatically by the external `cron-job.org` webhook — this is a GitHub UI labelling artefact, not a human action. “Scheduled” runs originate from GitHub’s own fallback cron trigger.

3. Data Acquisition & Feature Engineering

A predictive model is only as good as its features. The dataset combines meteorological features and ambient pollutants from two data sources.

3.1 Ambient Pollutants — AQICN API

Feature	Description	Unit
aqi	Overall Air Quality Index	AQI scale
pm25	Fine particulate matter	$\mu\text{g}/\text{m}^3$
pm10	Coarse particulate matter	$\mu\text{g}/\text{m}^3$
o3	Ozone concentration	ppb
no2	Nitrogen dioxide	ppb
so2	Sulfur dioxide	ppb
co	Carbon monoxide	ppm

3.2 Meteorological Conditions — Open-Meteo API

Feature	Description	Notes
temperature	Ambient temperature	Celsius
humidity	Relative humidity	%
wind_speed	Wind speed	km/h
wind_direction	Wind direction	Decomposed into sin/cos
pressure	Atmospheric pressure	hPa
cloud_cover	Cloud cover	%
visibility	Visibility	km

3.3 Engineered Temporal Features

- **Cyclic Time Features** — Sin/cos conversions of `hour`, `day_of_week`, and `month`, allowing models to learn diurnal, weekly, and seasonal cycles smoothly without ordinal discontinuities.
- **Lag Features** — 1-hour lag of AQI (`aqi_lag_1h`) and 1-hour lag of PM2.5 (`pm25_lag_1h`) to capture momentum.
- **Rate of Change** — AQI rate of change over the last hour (`aqi_change_rate`).
- **Rolling Average** — 3-hour rolling average (`rolling_aqi_mean_3h`) to capture short-term trends.

3.4 Target Variables

For each row at timestamp t , target variables are the future ground-truth AQI values:

$$y_{t+24h}, \quad y_{t+48h}, \quad y_{t+72h}$$

4. Model Training & Validation Protocol

To prevent temporal data leakage (predicting the past using future data), we avoid random k -fold cross-validation and instead use a **Chronological Time-Series Split**.

Split	Proportion	Purpose
Training Set	70% of chronological history	Model fitting
Validation Set	15% of chronological history	Hyperparameter selection
Test Set	15% of chronological history	Final evaluation

4.1 Candidate Models

Five models are trained and compared to handle multi-step forecasting:

- 1. Multi-Output Ridge Regression** — A linear model with L_2 regularisation acting as the baseline.
- 2. Multi-Output Random Forest Regressor** — An ensemble bagger capturing non-linear splits and feature interactions.
- 3. Multi-Output XGBoost Regressor** — A gradient-boosted ensemble optimised for structured tabular features.
- 4. Multi-Output LightGBM Regressor** — A fast, leaf-wise tree growth gradient booster with low memory overhead.
- 5. LSTM Neural Network** — A recurrent architecture built in TensorFlow to model sequential temporal relationships.

Model Selection Criterion

The training pipeline automatically tracks **Root Mean Squared Error (RMSE)** on the validation set across all models and saves the best-performing model artifact to the Local Model Registry.

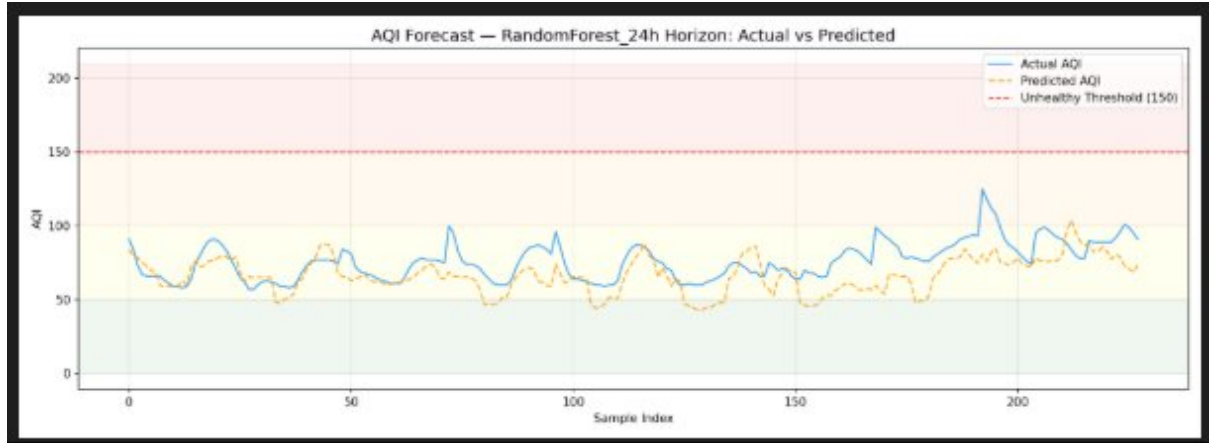


Figure 3: RandomForest model — AQI Forecast at the 24-hour horizon: Actual vs. Predicted. The dashed red line marks the Unhealthy threshold (AQI = 150). The model tracks real-world AQI dynamics closely across the test set.

5. Model Explainability (SHAP Analysis)

To ensure the model is not a “black box,” we integrate SHAP (SHapley Additive exPlanations) to explain individual forecasts and audit global feature importance.

5.1 Key Findings

- **Lag Features Dominate.** `aqi_lag_1h` and `rolling_aqi_mean_3h` yield the highest SHAP values, confirming that air quality exhibits high *inertia* — the next 24-hour AQI is heavily dependent on the current level.
- **Meteorological Drivers Emerge at Longer Horizons.** For the 48-hour and 72-hour forecasts, weather variables become significantly more prominent:
 - `temperature` — higher temperatures promote thermal convection, dispersing pollutants.
 - `wind_speed` — high winds dilute particle concentration.
- **Diurnal Cycles Captured.** The engineered hour components (`hour_sin` / `hour_cos`) successfully capture rush-hour spikes from morning and evening traffic emissions.

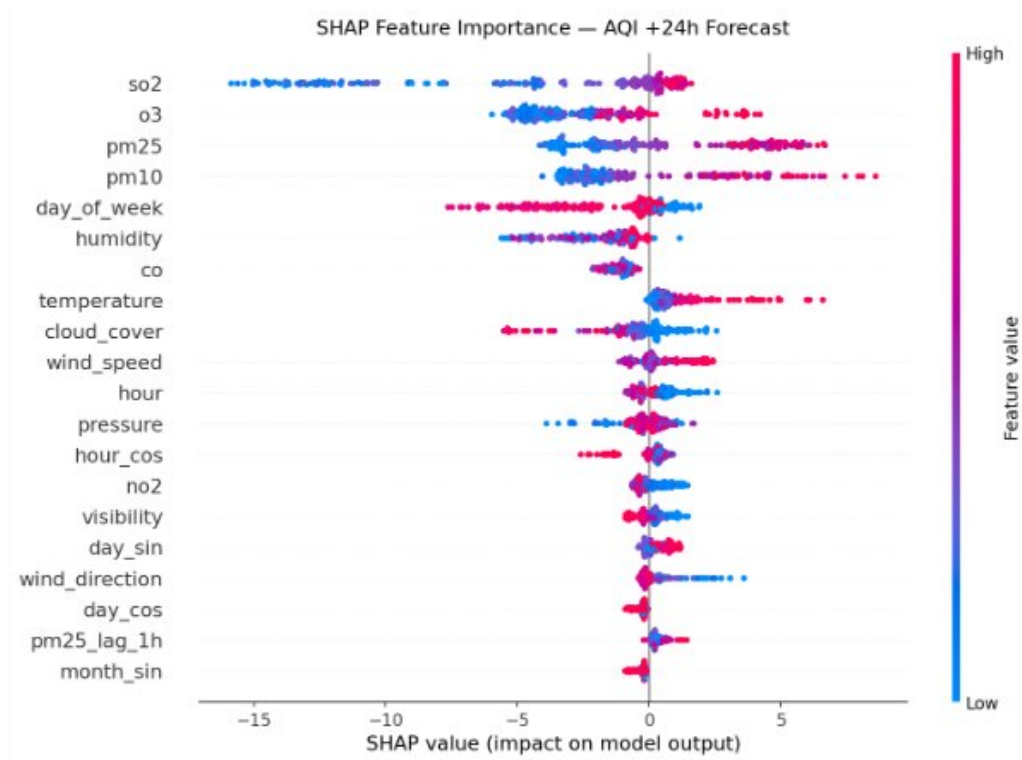


Figure 4: SHAP Feature Importance — AQI +24h Forecast. Each dot represents one prediction; colour indicates feature value (pink = high, blue = low). Features are ranked by mean absolute SHAP impact on the model output.

6. Serving and User Interface (Dashboard)

The predictions are served via an interactive Streamlit application that consumes the FastAPI backend and presents a rich, user-facing view of the system's output.

6.1 Dashboard Features

Component	Description
AQI Status Cards	Dynamic colour-coded cards for current AQI: Good , Moderate , Unhealthy , Hazardous
Pollutant Readings	Real-time measurements of all six primary pollutants (PM2.5, PM10, O ₃ , NO ₂ , SO ₂ , CO)
3-Day Forecast	24-hour, 48-hour, and 72-hour future AQI forecasts with integrated health warnings
SHAP Viewer	Embedded SHAP explainability plots for interpreting the current forecast

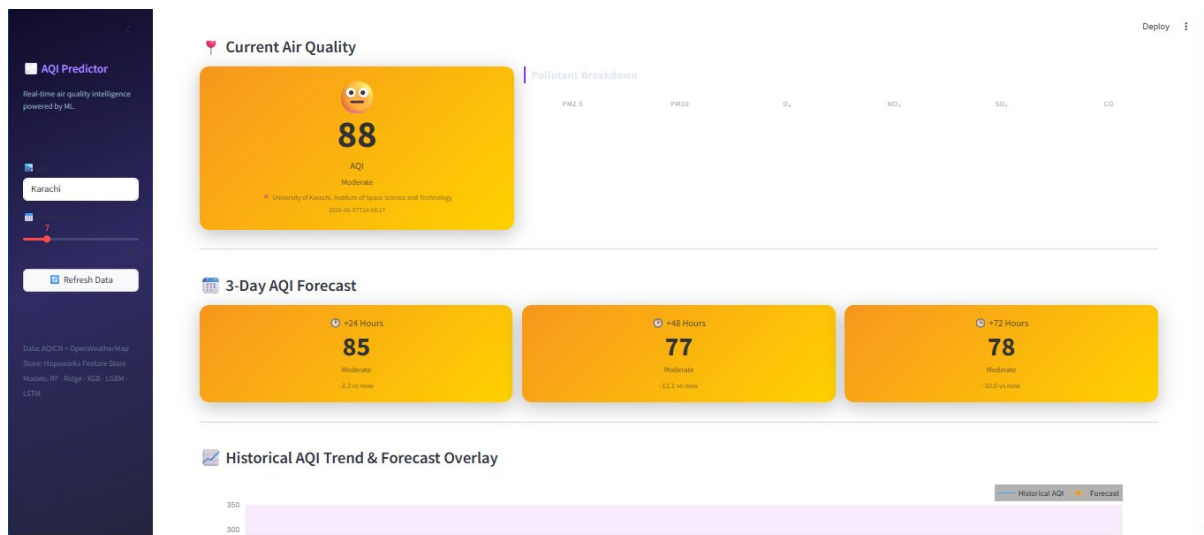


Figure 5: Streamlit AQI Predictor Dashboard — showing current AQI of 88 (Moderate) for Karachi, with 3-day forecasts of 85, 77, and 78 at the +24h, +48h, and +72h horizons respectively, alongside a Historical AQI Trend & Forecast Overlay chart.